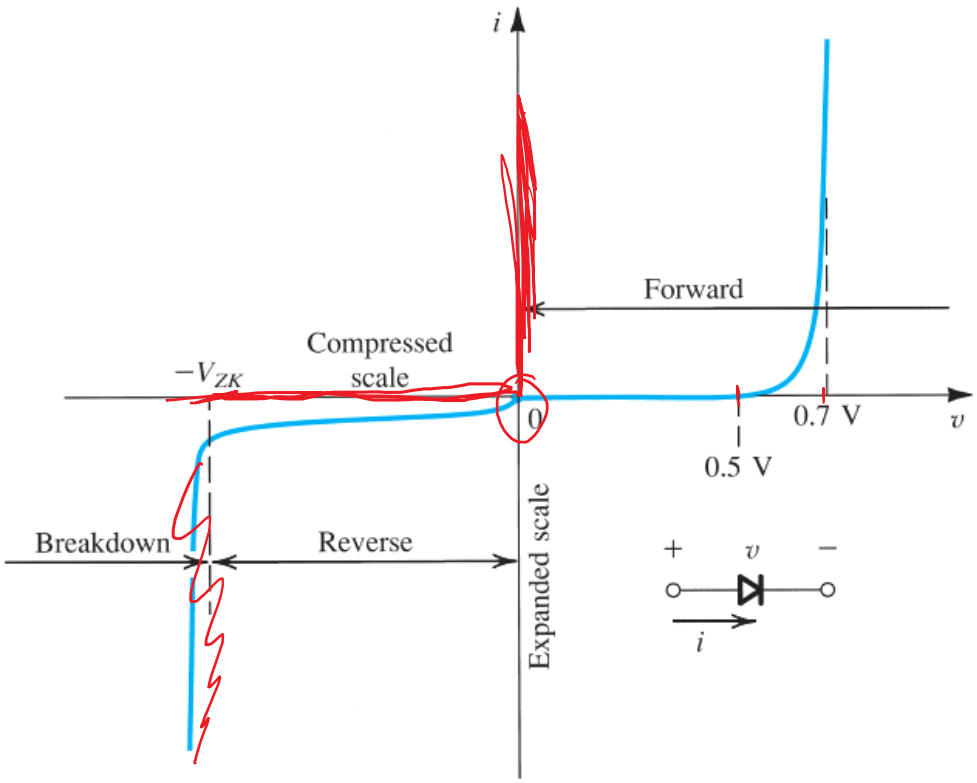
# Reading

Adel S. Sedra and Kenneth C. Smith, **Microelectronic Circuits** 7th Edition, *Oxford University Press*, 2014.

- Chapter 4.1, 4.3, 4.4

# Recap-1



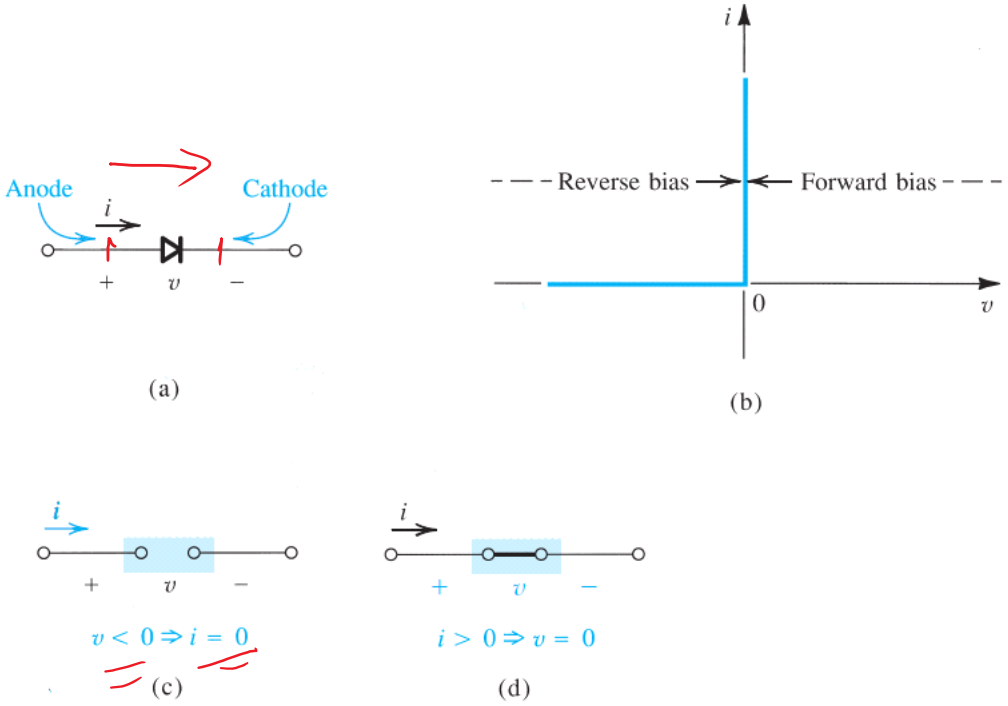
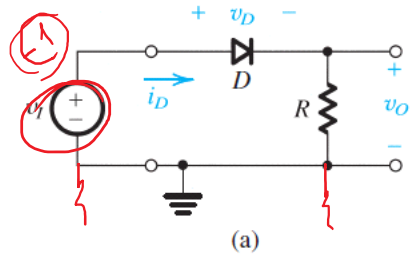
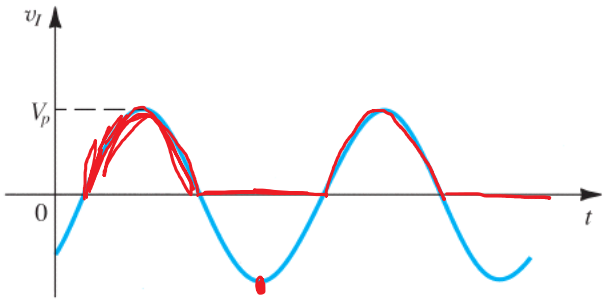




**Figure 4.1** The ideal diode: **(a)** diode circuit symbol; **(b)** $i$–$v$ characteristic; **(c)** equivalent circuit in the reverse direction; **(d)** equivalent circuit in the forward direction.

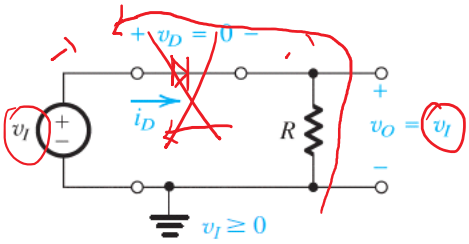
# Recap-2



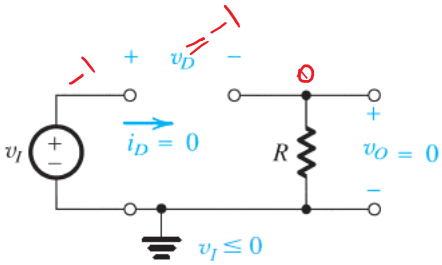

(a)



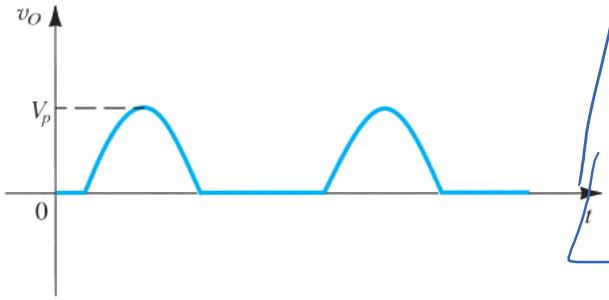

(b)



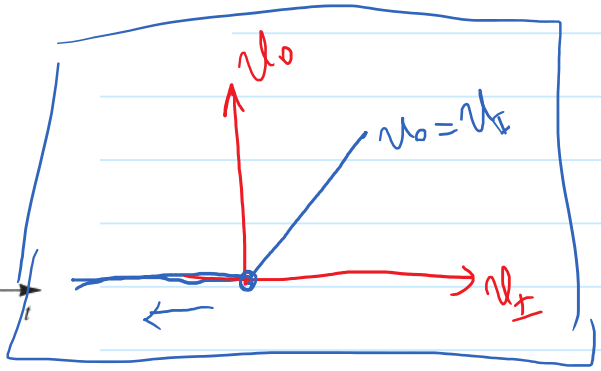

(c)

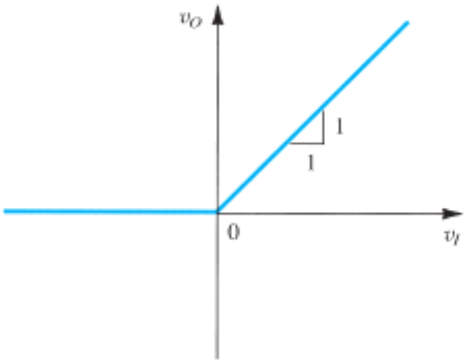



(d)

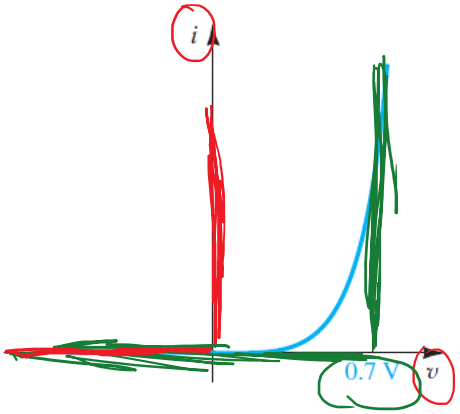



(e)

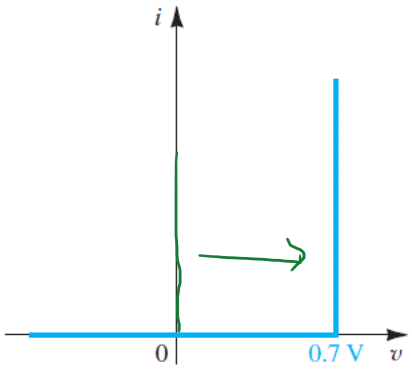


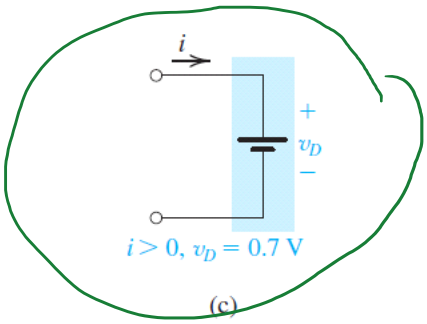
# Constant Voltage Drop Model



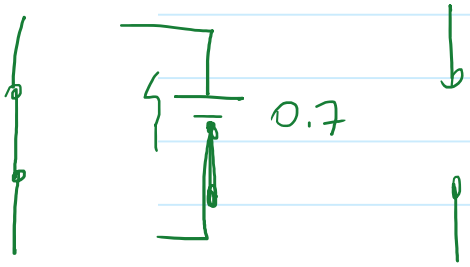

(a)



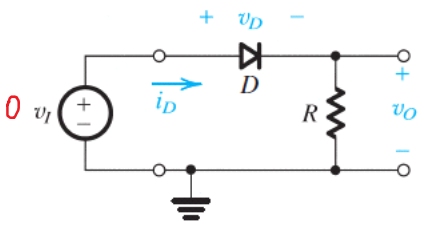

(b)



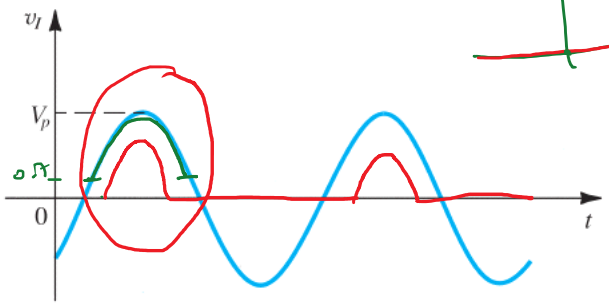

(c)



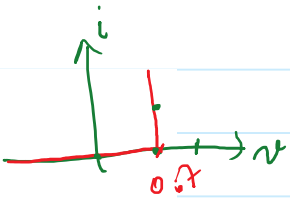
# Example

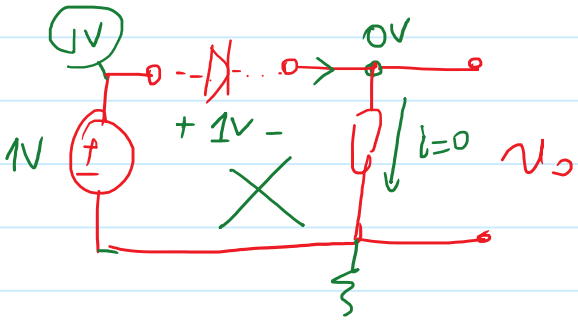

+ $v_D$ −
D
$i_D$
$v_I$
R
+
$v_O$
−

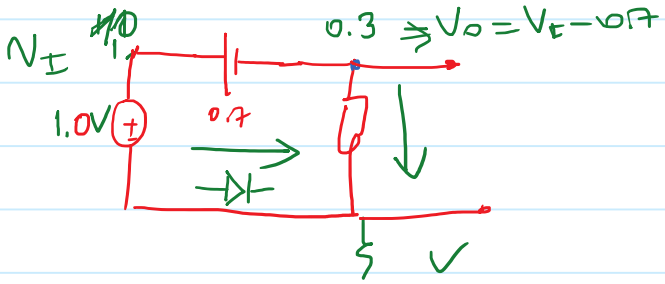
(a)

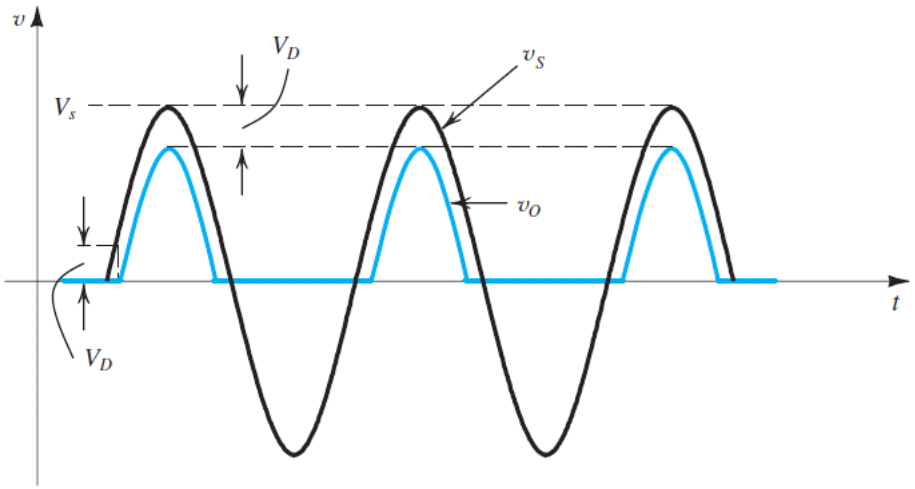

$v_I$
$V_p$
0
t

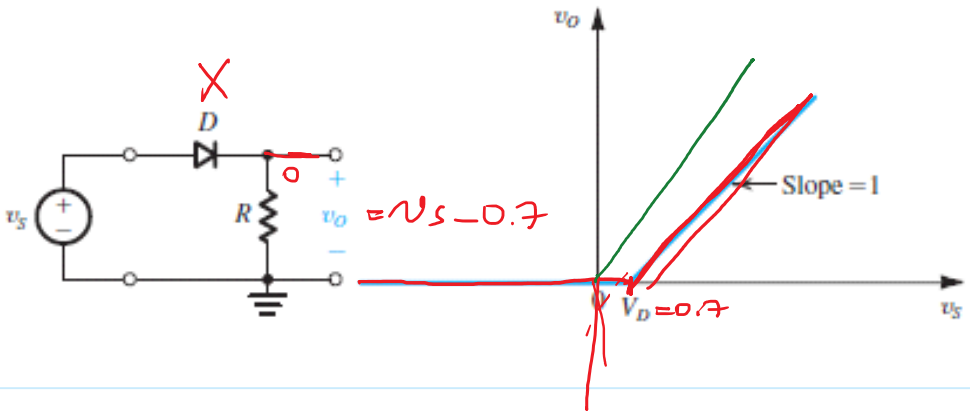
(b)


i
v
0.7

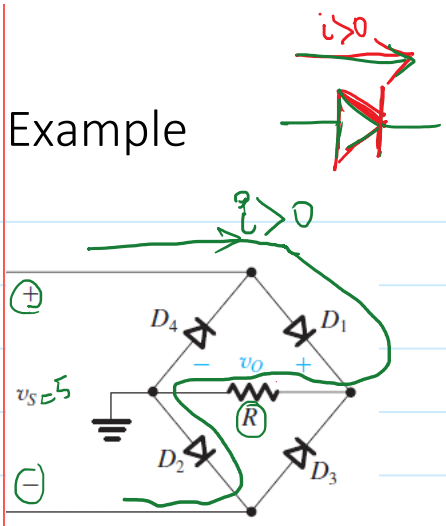

1V
0V
0
+ 1V −
1V
i=0
$v_o$

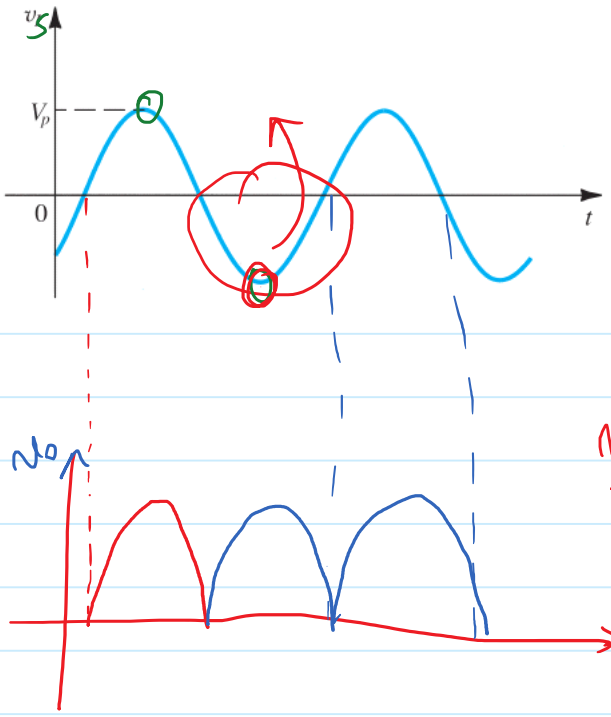

$V_I$
1.0V
0.7
0.3 ⇒ $V_o = V_I - 0.7$
V

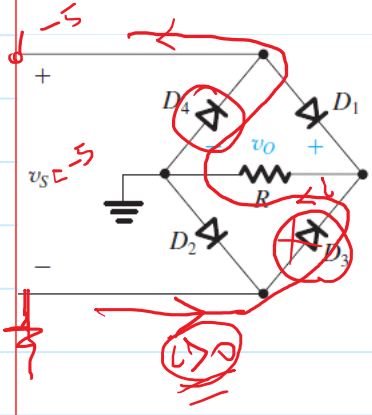

v
$V_D$
$v_S$
$V_s$
$v_O$
t
$V_D$

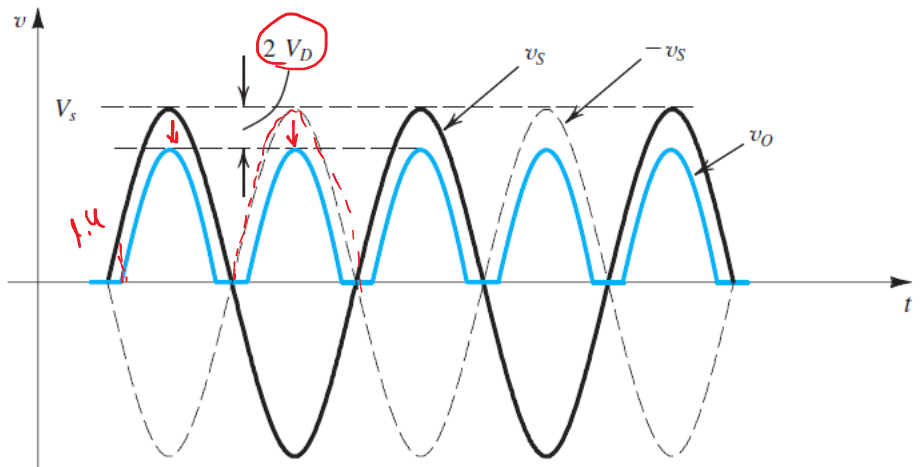

X
D
0
+
$v_S$
R
$v_O$
−
$= v_S - 0.7$
$v_O$
Slope = 1
$V_D = 0.7$
$v_S$

# Example

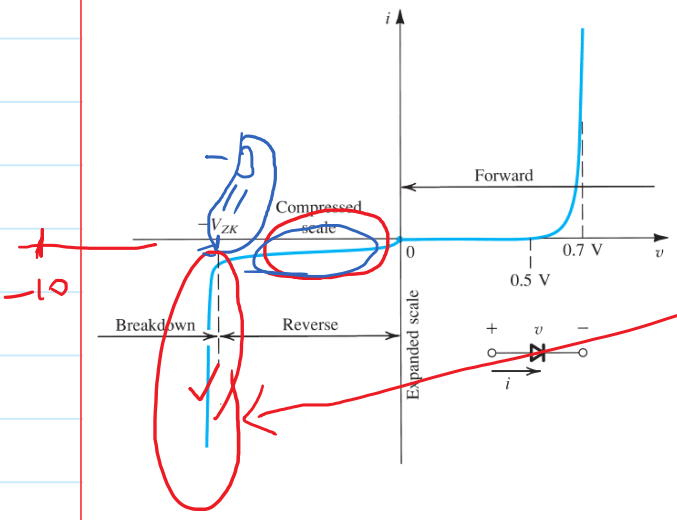


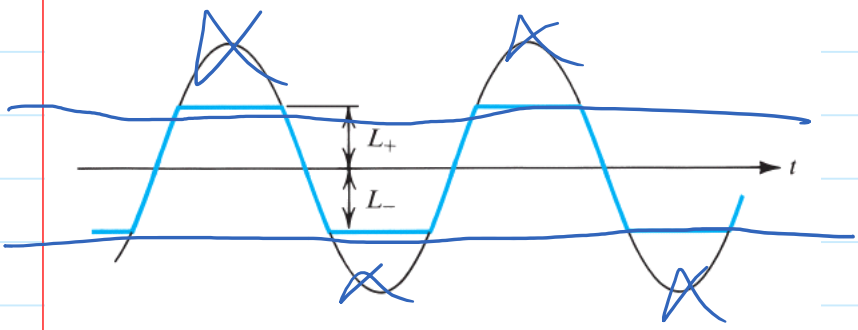



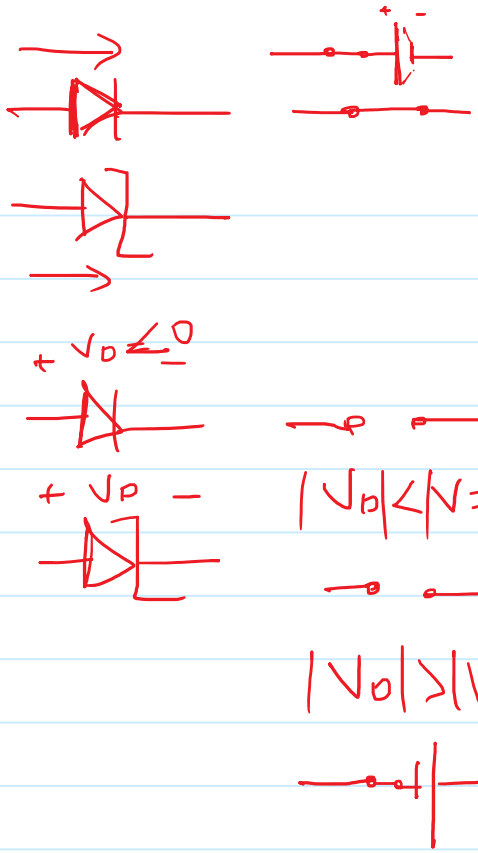



# Reverse Breakdown - Zener Diodes

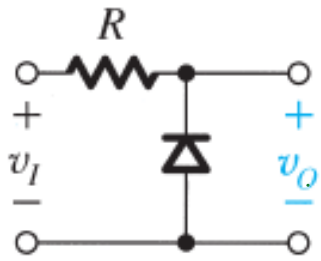


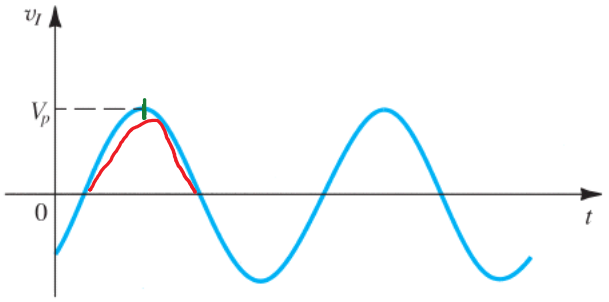
# Limiter Circuits



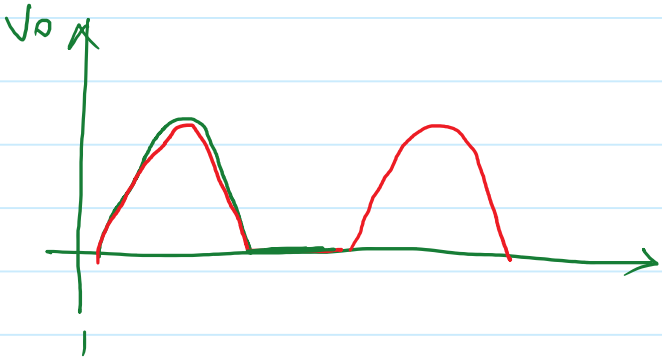
# Example

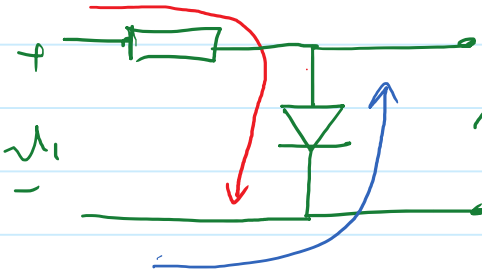


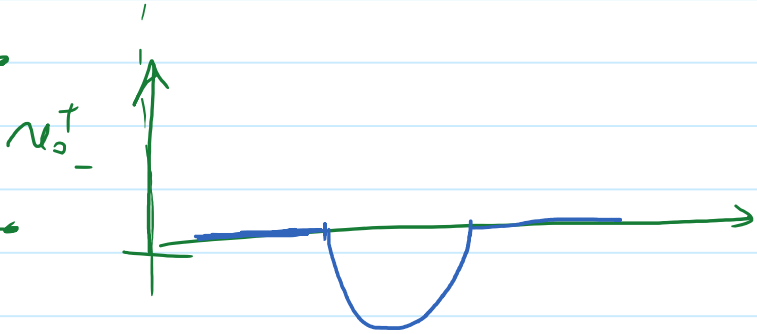



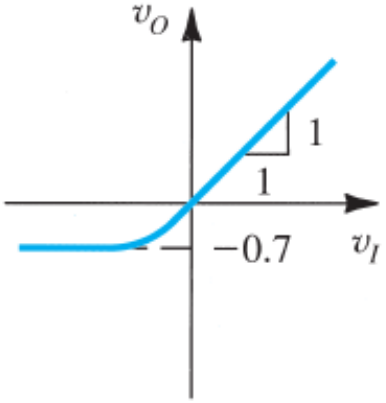




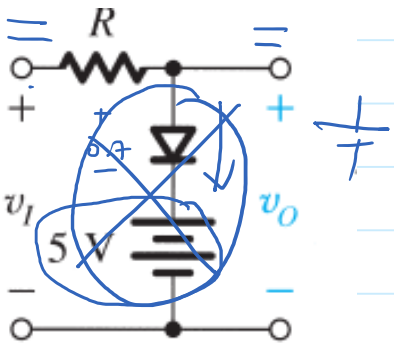


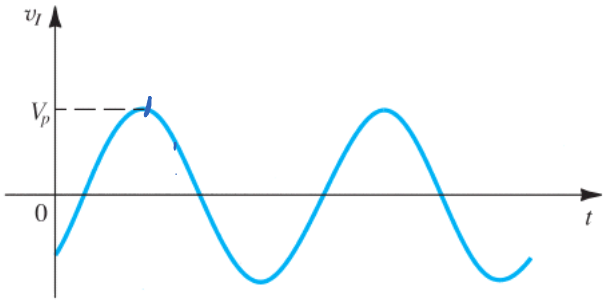



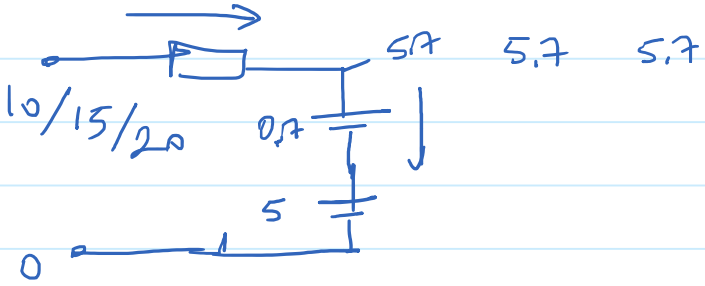



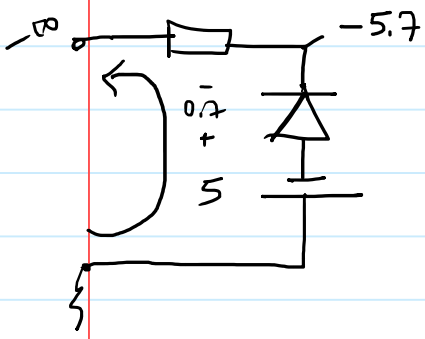



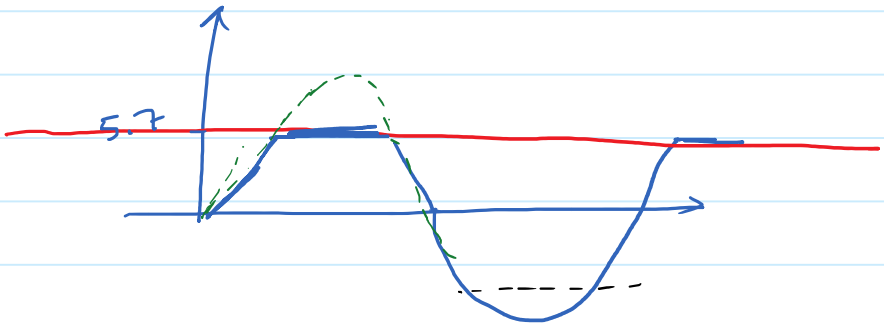
# Example

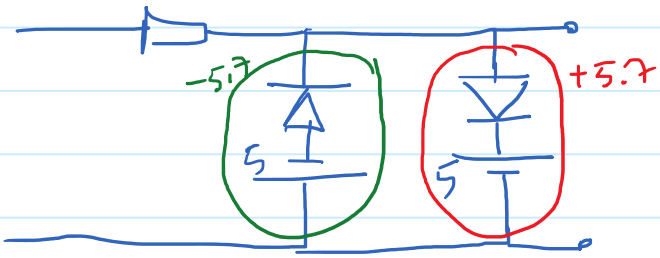


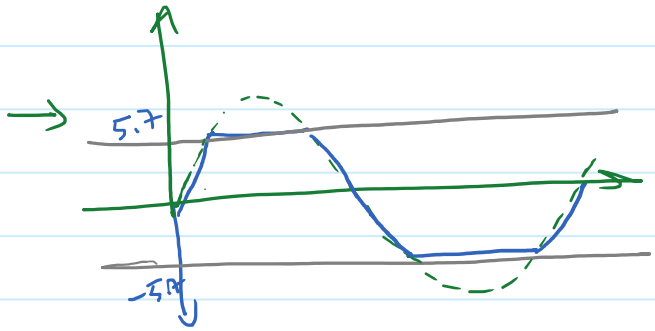



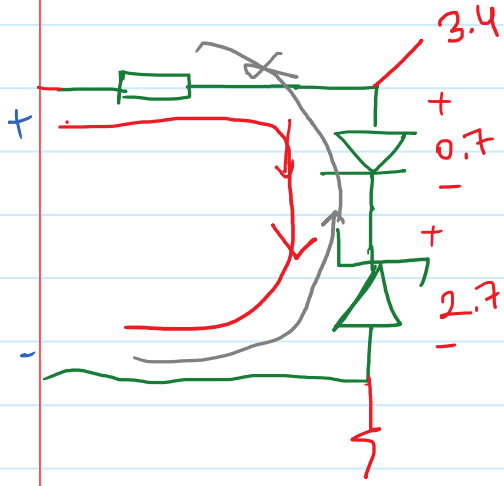



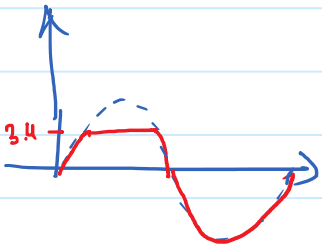



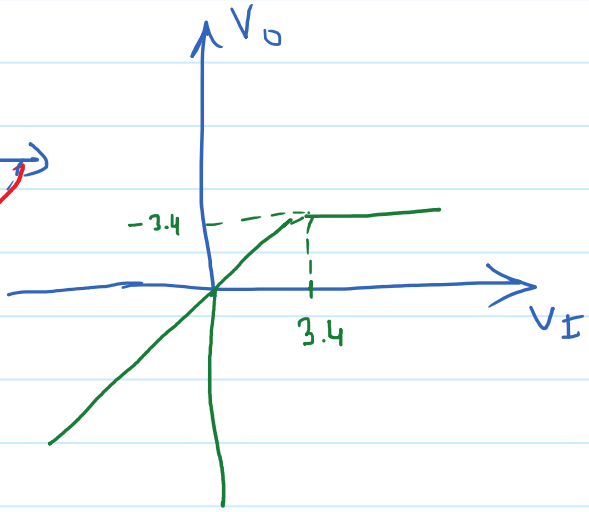



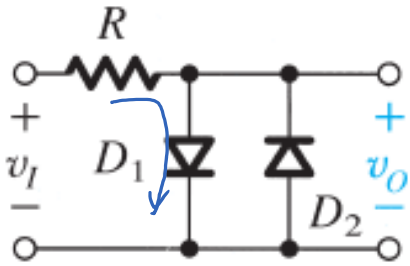

- Zener+

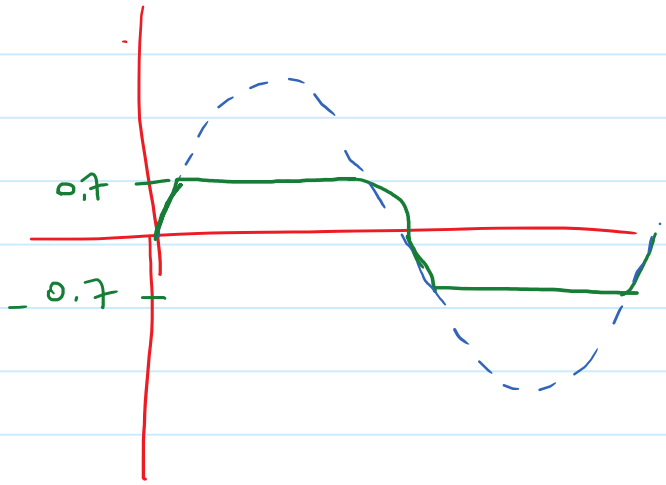


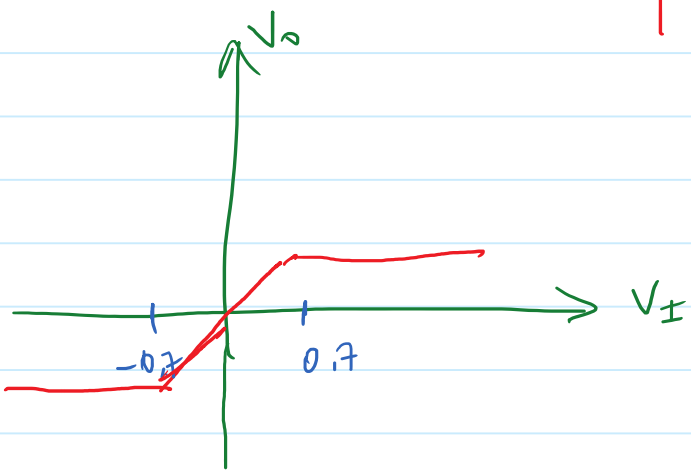



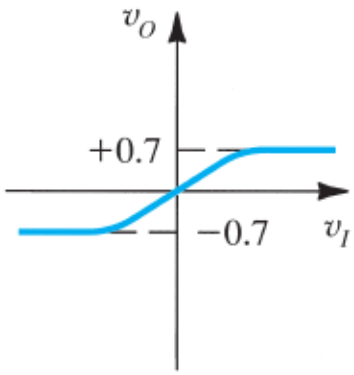



# Example

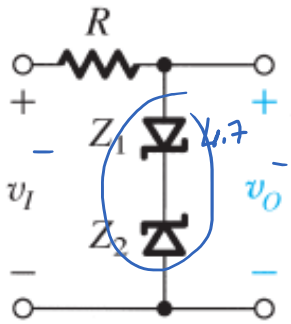


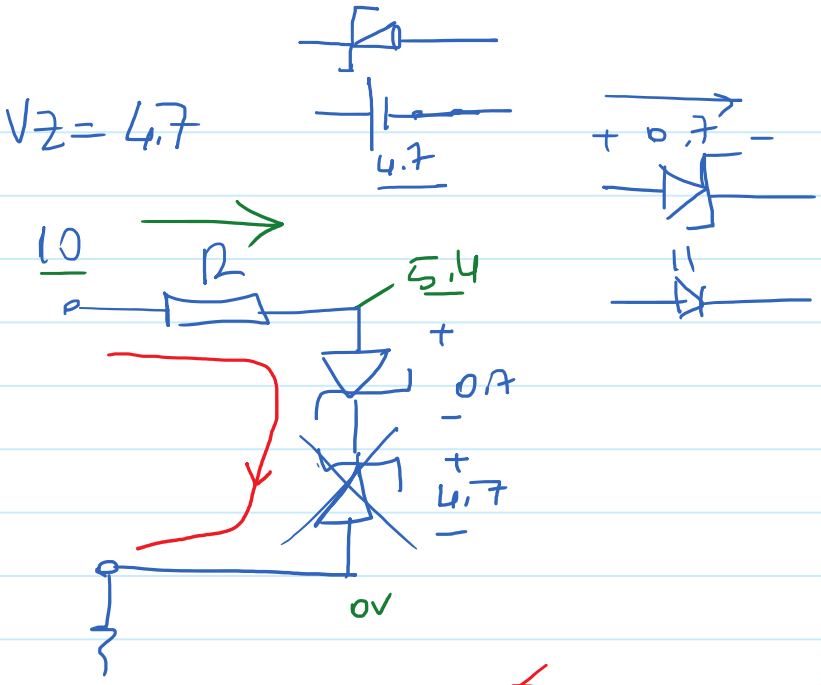



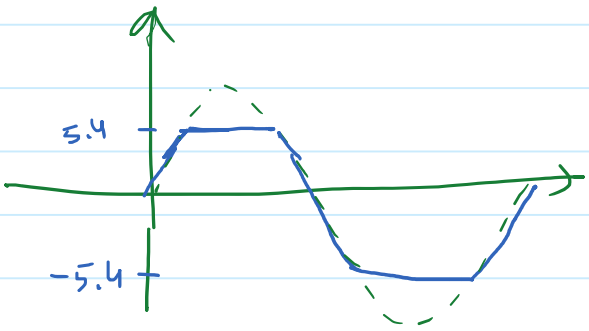



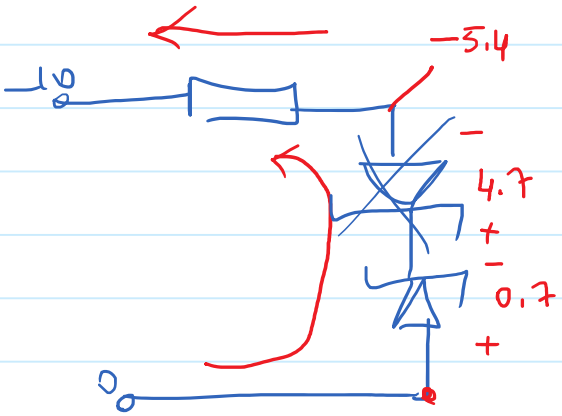



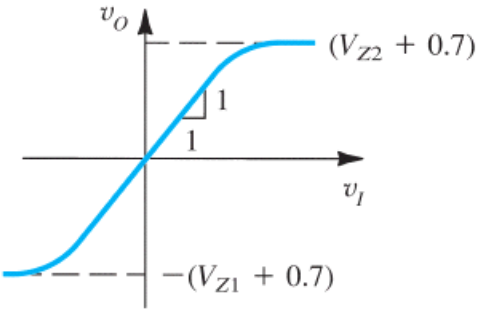
# Example


R
+
$v_I$
−
$Z_1$
4.7
$Z_2$
+
$v_O$
−

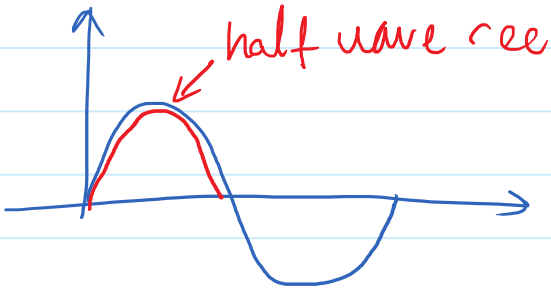

Vz = 4.7
4.7
+ 0.7 −
10
R
5.4
+
0.7
−
+
4.7
−
0V

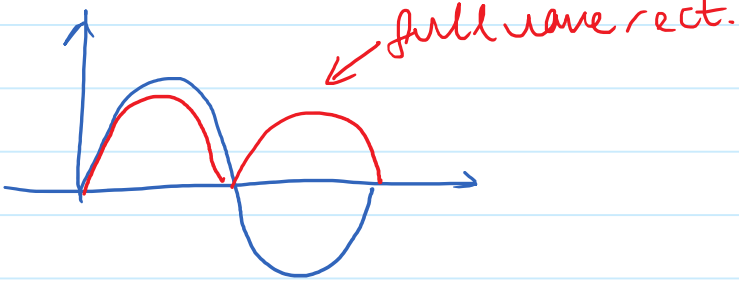

5.4
−5.4


−5.4
−10
−
4.7
+
−
0.7
+
0


$v_O$
$(V_{Z2} + 0.7)$
1
1
$v_I$
$-(V_{Z1} + 0.7)$

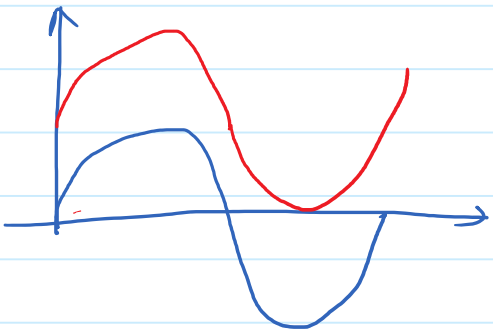
# Review



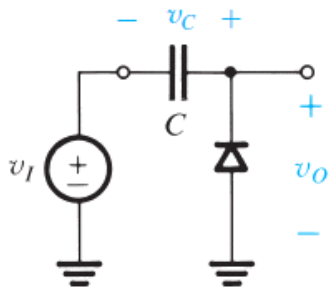



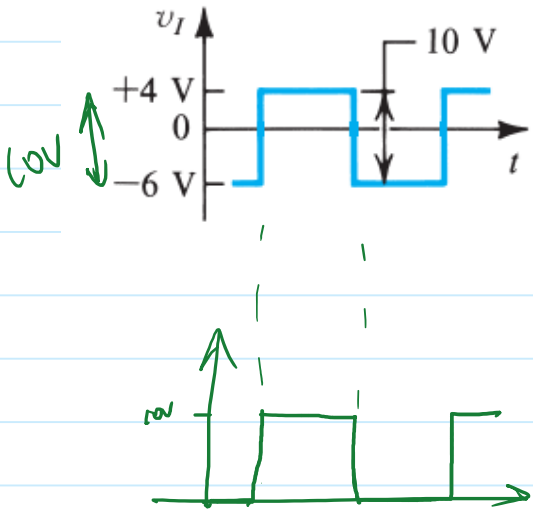






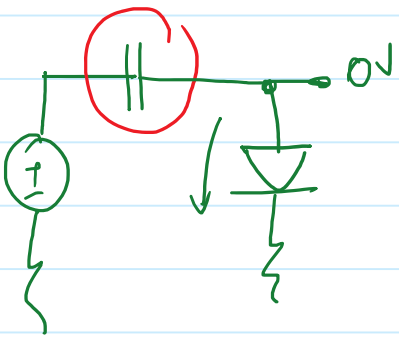
# Clamping Circuit

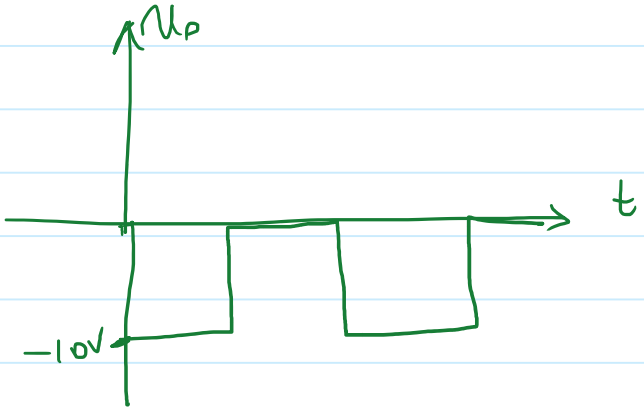


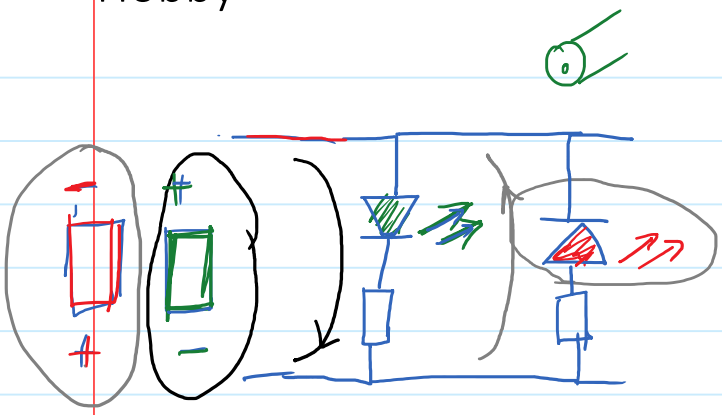



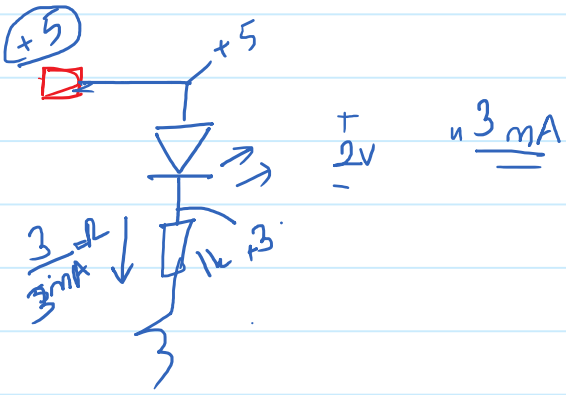









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3